



AI Infrastructure

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ML Pipeline Introduction

Objectives

- Explain the motivation behind how we came to AI Infrastructure components. A lot of these components may have correlation with traditional ML infrastructure.
- Explain the components of an ML infrastructure, including data pipelines, model training environments, and deployment platforms.
- Implement MLOps practices for version control, monitoring, and reproducibility.



Introduction to ML Modeling

The ML model management and the delivery of highly performing model is as important as the initial build of the model by choosing right dataset.

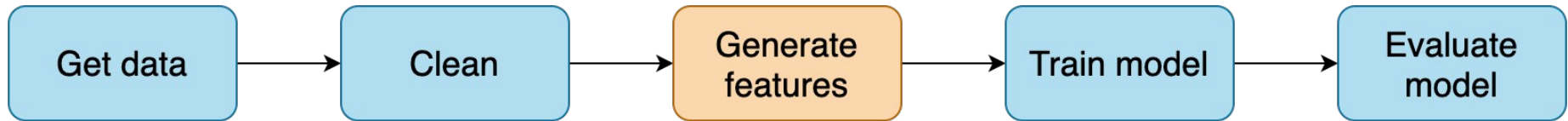
The concepts around model retraining, model versioning, model deployment and model monitoring are the basis for machine learning operations (MLOps) that helps the data science teams deliver highly performing models.

ML and AI are growing

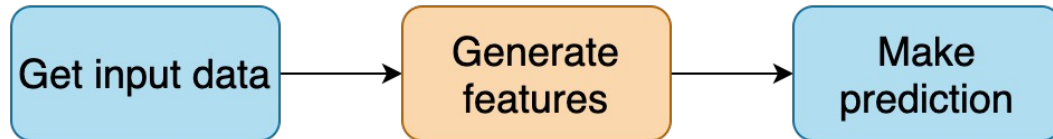
- The use of Machine Learning (ML) and Artificial Intelligence (AI) has increased substantially in enterprise data analytics scenarios to extract valuable insights from the business data.
- Hence, it is very important to have an ecosystem to build, test, deploy, and maintain the enterprise grade machine learning models in production environments.

ML Model Pipelines

Model training



Model serving (batch/online)



ML Development

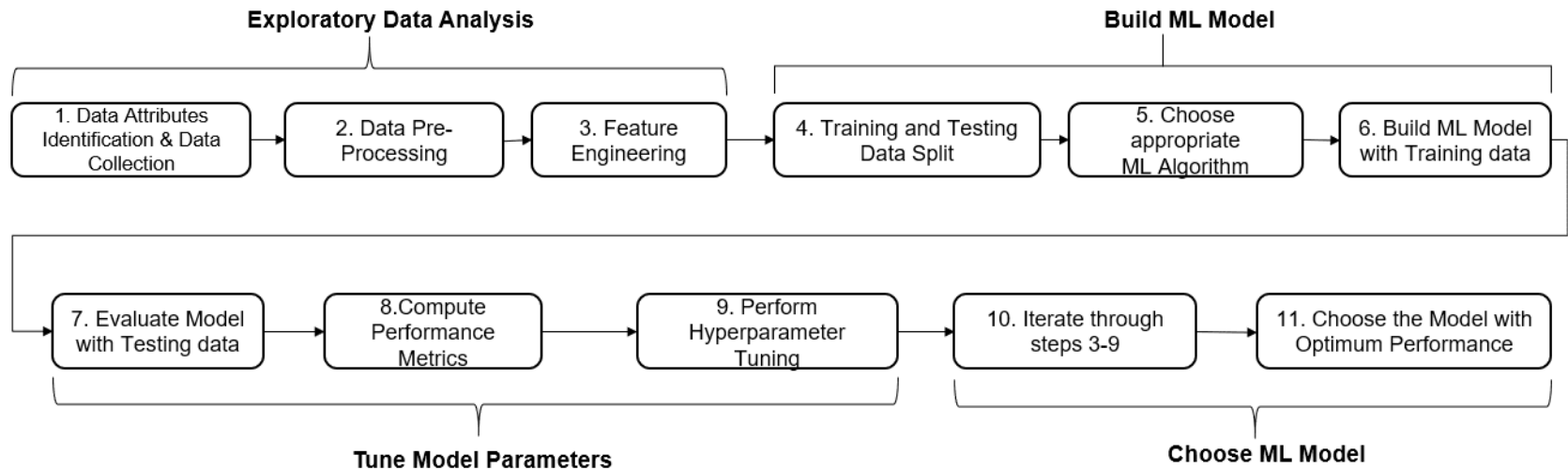
- The ML model development involves
 - data acquisition from multiple trusted sources,
 - data processing to make suitable for building the model,
 - choose algorithm to build the model,
 - build model,
 - compute performance metrics and
 - choose best performing model.

Model Maintenance

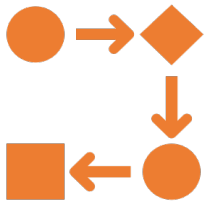
- The model maintenance plays critical role once the model is deployed into production.
 - The maintenance of machine learning model includes keeping the model up to date and relevant in tune with the source data changes as there is a risk of model becoming outdated in course of time.
 - The configuration management of ML model play an important role in model management as the number of models grow.

Model Development

- **Machine Learning (ML) Model Lifecycle** refers to the process that covers right from source data identification to model development, model deployment and model maintenance. At high level, the entire activities fall under two broad categories, such as ML Model Development and ML Model Operations.

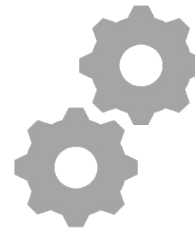


Exploratory Data Analysis



starts once business hypothesis is ready.

This step takes 40-50% of total project time as the model outcome depends on the quality of input data being fed to train the model.



Involves data attributes identification, data preprocessing and feature engineering.

Attributes' identification involves identification of predictor/features variables (inputs) and target/class variable (output), along its data types (string or numeric or datetime) and classification of features into categorical and continuous variables that helps in applying appropriate treatment to be given to the variable by the algorithm while building the model.

Data pre-processing involves identification of missing values and outliers and fill these gaps by computing mean or median for quantitative attributes and mode for qualitative attributes of data to improve the predictive power of model.

The outliers cause increased mean and standard deviation, that can be eliminated by taking natural log value which reduces the variation caused by extreme values.

Feature Engineering

... is the process of using [domain knowledge](#) of the data to create [features](#) that make [machine learning](#) algorithms work.

Feature engineering is fundamental to the application of machine learning, and is both difficult and expensive. The need for manual feature engineering can be obviated by automated [feature learning](#).

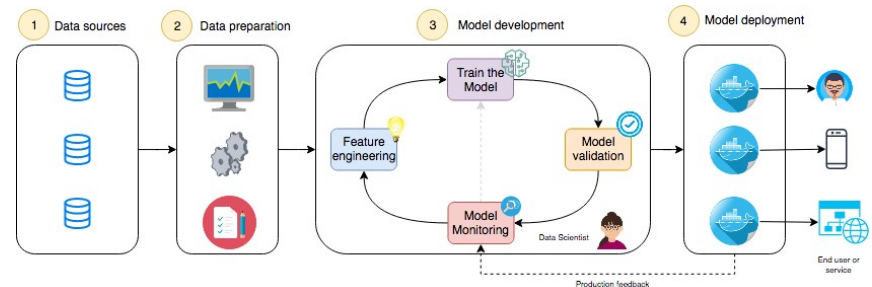
- the raw dataset is processed to convert data types of string or datetime or numeric ones to numeric vectors for an ML algorithm to understand and build efficient predictive model.

From the definition above, I want to highlight the following points:

- Feature engineering is expensive and time-consuming because it depends on the problem and domain knowledge.
- Feature engineering can be a manual process but in some cases...
- Feature engineering process can be automated.

Feature Engineering

- Also, the labelled categorical data (e.g., color → red, green, blue; performance → poor, fair, good, very good, excellent; risk → low, medium, high; status → started, in-progress, on-hold, closed; gender → male/female; is_loan_approved/is_claim_fraudulent → yes/no) cannot be understood by an ML algorithm in its true context, hence it is required to be converted into numeric data.
- Feature Encoding is a technique to transform the categorical data into continuous (numerical) values, e.g., Encode color as 1,2,3; performance as 1,2,3,4,5; risk as 1,2,3; status as 1,2,3,4; gender / is_loan_approved / is_claim_fraudulent as 0/1 etc.
 - It is recommended to use label encoding in case categorical variables have no ordered relationship (e.g., color and status), ordinal encoding in case categorical variables have an ordered relationship (e.g., performance, risk) and one hot encoding in case categorical variable data is binary in nature (e.g., is_loan_approved, is_claim_fraudulent)



Feature Engineering #2

- A set of libraries are available in R or Python to implement these encoding methods.
- In some cases, a set of dummy variables or derived variables are created, especially in handling 'date' data types.
- Once the categorical text data is converted into numeric data, the data is ready to be fed to the model;
- As a last step, it is required to choose the appropriate features that help in improving the accuracy of model, by using the techniques such as Univariate Selection (statistical measure), Feature Importance (model property) and Correlation Matrix (identifies which features are most related to the target variable).
- These methods detect Collinearity between two variables where they are highly correlated and contain similar information about the variance within a given dataset. One may find the Variance Inflation Factor (VIF) useful to detect Multicollinearity where highly correlated three or more variables are included in the model.

Exploratory Data Analysis

